

Virtual Welfare

Imagine a single buyer will arrive with their private value v , drawn from a known continuous distribution F . We want to design DSIC mechanisms.

What mechanism should you use to maximize *welfare*?

What should you do to maximize (expected) revenue?

Definition 1. In a deterministic mechanism, given other bids $\mathbf{b}_{-i}$, bidder i 's *critical bid* is the minimum bid $b_i^* = \min\{b_i : x_i(b_i, \mathbf{b}_{-i}) = 1\}$ such that bidder i is allocated to.

Then with $\mathbf{b}_{-i}$ fixed, for all winning $v_i \geq b_i^*$, i 's payment $p_i(v_i, \mathbf{b}_{-i}) = b_i^*$ is their critical bid.

What is winner i 's critical bid in a single-item auction?

What about in the k identical item setting?

Revenue Maximization and Myersonian Virtual Welfare

Recall:

- The revelation principle says that it's without loss to focus only on truthful mechanisms.
- Payment is determined by the allocation:

$$p_i(b_i, \mathbf{b}_{-i}) = b_i \cdot x_i(b_i, \mathbf{b}_{-i}) - \int_0^{b_i} x_i(z, \mathbf{b}_{-i}) dz$$

For this lecture, values are independent and nonnegative. Each F_i has a density f_i , positive on the interior of its support, and finite expected value. Payments are normalized so that a bidder of value zero pays zero.

Fixed, choice, and randomness. For the expected-payment calculation and the designer's optimization:

- What is fixed?
- What can the designer choose?
- What are we averaging over?

Probability for the revenue calculation

If V is nonnegative with density f and CDF F , then

$$\mathbb{E}[g(V)] = \int_0^\infty g(t)f(t) dt, \quad \Pr[V \geq t] = 1 - F(t) = \int_t^\infty f(s) ds.$$

The first formula averages any integrable function $g(V)$, not just V itself. For a continuous distribution, including the endpoint does not change a probability.

Changing the order of integration. The region $0 \leq s \leq t$ can be described in two ways: fix t and let s run from 0 to t , or fix s and let t run from s to ∞ . For nonnegative h ,

$$\int_0^\infty \int_0^t h(s, t) ds dt = \int_0^\infty \int_s^\infty h(s, t) dt ds.$$

Both sides add up the same contributions. The inner integral is evaluated first.

Averaging in stages. For independent V, W , first fix $W = w$ and average $H(V, w)$ over V , then average the resulting function over W :

$$\mathbb{E}_W[\mathbb{E}_V[H(V, W)]] = \mathbb{E}_{V, W}[H(V, W)].$$

Independence keeps the distribution of V unchanged when W is fixed. Here $V = v_i$ and $W = \mathbf{v}_{-i}$. Also, $\mathbb{E}[\sum_i Z_i] = \sum_i \mathbb{E}[Z_i]$ by linearity, without needing independence.

We want to maximize $\mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[\sum_i p_i(\mathbf{v})]$.

$$\mathbb{E}_{v_i \sim F_i}[p_i(v_i, \mathbf{v}_{-i})] =$$

where

$$\varphi_i(v_i) =$$

is the Myersonian virtual value and (*) follows by switching the order of integration. Then

$$\text{REVENUE} = \mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[\sum_i p_i(\mathbf{v})] =$$

$$= \text{VIRTUAL WELFARE}$$

Given this conclusion, how should we design our allocation rule x to maximize expected virtual welfare (expected revenue)?

When would this cause a problem with incentive-compatibility?

Definition 2. A distribution F is regular if the corresponding virtual valuation function $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ is strictly increasing.

Suppose we are in the single-item setting and all of the distributions are regular. What do the payments look like in the virtual-welfare-maximizing allocation?

For a fixed $\mathbf{b}_{-i}$, if i is the winner, then i 's payment is i 's critical bid, which is

Exercise: what about for k identical items?

Claim 1. A virtual welfare maximizing allocation x is monotone if the virtual value functions are regular, with a fixed priority order for breaking ties.

Exercise: Argue this.

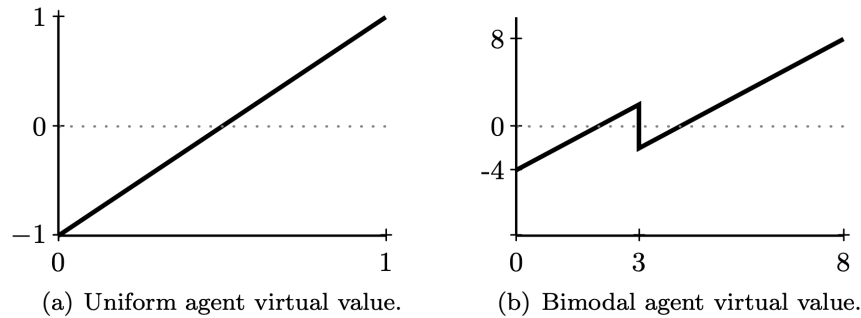


Figure 1: Virtual value functions $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ for the uniform and bimodal agent examples.

It will be helpful to keep the following two examples in mind:

- a. a uniform agent with $v \sim U[0, 1]$. Then $F(x) = x$ and $f(x) = 1$.
- b. a bimodal agent with

$$v \sim \begin{cases} U[0, 3] & w.p. \frac{3}{4} \\ U(3, 8] & w.p. \frac{1}{4} \end{cases} \quad \text{and} \quad f(v) = \begin{cases} \frac{1}{4} & v \in [0, 3] \\ \frac{1}{20} & v \in (3, 8] \end{cases}$$

Do the following:

- Calculate the virtual values for both examples.
- Are they regular? Are there any issues using the allocation that maximizes expected virtual welfare?
- What does that allocation actually look like?

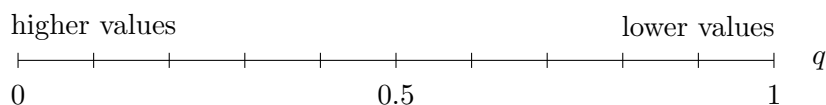
Quantile Space

Position in the distribution. Assume F is continuous and strictly increasing on its support. An agent's *quantile* is the fraction of the distribution with higher values: $q = 1 - F(v)$. We let $v(q)$ denote the corresponding value.

Consider a distribution that is $U[\$0, \$10]$. A value of \$9 is in the top 10%, so its quantile is 0.1 and $v(0.1) = \$9$. What quantile corresponds to a value of \$2?

Quantile distribution. Let $V \sim F$ and let $Q = 1 - F(V)$ be its quantile. What is the probability that a random draw is in the top 20%? Between the top 20% and the top 50%?

What probability belongs to each interval on this quantile axis?



$$\Pr[Q \leq a] = \underline{\hspace{2cm}} \quad (0 \leq a \leq 1), \quad Q \sim \underline{\hspace{2cm}}.$$

Averaging. For a quantity $g(v)$, its value at quantile q is $g(v(q))$. Rewrite the expectation in quantile space:

$$\mathbb{E}[g(V)] = \int_0^\infty g(v)f(v) dv = \underline{\hspace{4cm}}.$$

What happened to the density factor?

Revenue. If we post price $v(q)$, what is the probability of a sale? What is the expected revenue $R(q)$?

$$\text{Sale probability} = \underline{\hspace{2cm}}, \quad R(q) = \underline{\hspace{2cm}}.$$

For $U[\$0, \$10]$, posting \$9 gives expected revenue $\underline{\hspace{2cm}}$.

Now draw the posted price from F , independently of the buyer's value. Write its expected revenue as an integral of R :

$$\text{Expected revenue} = \underline{\hspace{4cm}}.$$